

CMI Entrance Examination

Mock Test - Part A

Instructions: This section consists of 10 multi-part objective questions. For "Fill in the blank" questions, write your final numerical or algebraic answer in the space provided. For "True/False" questions, clearly write either True or False.

Question 1

4 points

Let $I_n = \int_0^{\pi/2} \sin^n(x) dx$ for integers $n \geq 0$.

1.1 (Fill in the blank) The exact numerical value of I_4 is _____.

1.2 (True/False) The sequence defined by $A_n = nI_n I_{n-1}$ is strictly decreasing for all $n \geq 1$. _____

1.3 (Fill in the blank) The exact value of the limit $\lim_{n \rightarrow \infty} \sqrt{n} I_n$ is _____.

1.4 (Fill in the blank) Let $S = \sum_{n=1}^{\infty} (-1)^n I_n$. The exact numerical value of S is _____.

Question 2

4 points

Consider the sequence of integers defined by $a_n = 2^{2n+1} - 2^{n+1} + 1$ for $n \geq 1$.

2.1 (True/False) For all integers $n \geq 1$, a_n is mathematically guaranteed to be a prime number. _____

2.2 (Fill in the blank) The greatest common divisor $\gcd(a_{10}, a_{11})$ is exactly _____.

2.3 (Fill in the blank) The total number of positive integers $n \leq 100$ such that $a_n \equiv 0 \pmod{5}$ is _____.

2.4 (Fill in the blank) The remainder when a_{2026} is strictly divided by 7 is _____.

Question 3

4 points

Let $p_k(n)$ denote the number of integer partitions of n into exactly k strictly positive parts.

3.1 (Fill in the blank) The exact integer value of $p_3(8)$ is _____.

3.2 (True/False) For any n and k , the number of partitions of n into *at most* k parts is mathematically identical to the number of partitions of n into parts of size *at most* k . _____

3.3 (Fill in the blank) The coefficient of x^6 in the formal power series expansion of $(1-x)^{-1}(1-x^2)^{-1}(1-x^3)^{-1}$ is _____.

3.4 (Fill in the blank) The number of partitions of 15 into distinct odd parts is exactly _____.

Question 4

4 points

Let $\triangle ABC$ be an acute-angled triangle with side lengths a, b, c , circumradius R , and inradius r . Let O be its circumcenter and H be its orthocenter.

4.1 (True/False) The perpendicular distance from the circumcenter O to the side BC is exactly $R \cos A$. _____

4.2 (True/False) The square of the distance between the orthocenter and the circumcenter can be universally expressed as $OH^2 = 9R^2 - (a^2 + b^2 + c^2)$. _____

4.3 (Fill in the blank) If $a = 5, b = 6$, and $c = 7$, the exact numerical distance OH is _____.

4.4 (Fill in the blank) The strict maximum possible value of the ratio $\frac{r}{R}$ for any valid triangle is _____.

Question 5

4 points

Let $P(x) = x^5 - 5x^3 + 4x - 2$. Let its five complex roots be r_1, r_2, r_3, r_4 , and r_5 .

5.1 (Fill in the blank) The exact number of strictly real roots of $P(x)$ is _____.

5.2 (Fill in the blank) The exact integer value of the symmetric sum $\sum_{i=1}^5 r_i^4$ is _____.

5.3 (True/False) The polynomial $P(x)$ is irreducible over the field of rational numbers $\mathbb{Q}$. _____

5.4 (Fill in the blank) The exact integer value of the product $\prod_{i=1}^5 (r_i^2 - 1)$ is _____.

Question 6

4 points

Let $z_1, z_2, \dots, z_{11}$ be the 11 distinct roots of the equation $z^{11} - 1 = 0$, where $z_1 = 1$. Let S be the set of these roots on the complex plane.

6.1 (Fill in the blank) The exact numerical value of the sum $\sum_{k=1}^{11} z_k^7$ is _____.

6.2 (Fill in the blank) The geometric product of the lengths of all 10 line segments connecting z_1 to the other roots z_k ($k \neq 1$) evaluates exactly to _____.

6.3 (True/False) There exist three distinct roots in S that can form the vertices of an equilateral triangle. _____

6.4 (Fill in the blank) Expressed in terms of the sine function, the maximum possible area of a triangle formed by choosing three distinct roots from S is _____.

Question 7

4 points

Consider the finite trigonometric series defined as $f(x) = \sin(x) + \sin(3x) + \sin(5x) + \dots + \sin(19x)$.

7.1 (Fill in the blank) For $x \neq k\pi$, a closed-form expression for $f(x)$ is strictly $\frac{\sin^2(Ax)}{\sin(x)}$. The integer value of A is _____.

7.2 (Fill in the blank) The exact numerical value of $f\left(\frac{\pi}{20}\right)$ is _____.

7.3 (True/False) The equation $f(x) = 0$ possesses exactly 9 distinct valid solutions within the open interval $(0, \pi)$. _____

7.4 (Fill in the blank) The exact value of the definite integral $\int_0^\pi f(x) \sin(x) dx$ is _____.

Question 8

4 points

Let M_n be an $n \times n$ tridiagonal matrix with 2s strictly on the main diagonal, and -1 s on both the super-diagonal and sub-diagonal.

8.1 (Fill in the blank) The exact integer value of the determinant of M_4 is _____.

8.2 (True/False) For any integer $n \geq 1$, the determinant of M_n is equal to exactly $n + 1$. _____

8.3 (Fill in the blank) Let v be the 5×1 column vector $(1, 2, 3, 4, 5)^T$. The sum of all 5 components of the resulting vector $M_5 v$ is _____.

8.4 (True/False) The matrix M_n is strictly positive definite for all integers $n \geq 1$. _____

Question 9

4 points

Let $x_1 = 3$, and define a real sequence recursively by $x_{n+1} = \frac{1}{2} \left(x_n + \frac{7}{x_n} \right)$ for all $n \geq 1$.

9.1 (True/False) The sequence (x_n) is strictly decreasing for all $n \geq 1$. _____

9.2 (Fill in the blank) The exact value of the limit $L = \lim_{n \rightarrow \infty} x_n$ is _____.

9.3 (Fill in the blank) The auxiliary sequence defined by $y_n = \frac{x_n - \sqrt{7}}{x_n + \sqrt{7}}$ strictly satisfies the power relation $y_{n+1} = (y_n)^k$. The integer value of k is _____.

9.4 (True/False) The infinite series $\sum_{n=1}^{\infty} (x_n - \sqrt{7})$ converges. _____

Question 10**4 points**

Consider a standard uncolored 8×8 grid of squares (resembling a chessboard).

10.1 (Fill in the blank) The maximum number of non-attacking rooks that can be placed simultaneously on the grid is _____.

10.2 (True/False) It is geometrically possible to tile the entire 8×8 grid using exactly 21 straight 1×3 trominoes and one 1×1 square tile placed exactly in the center of the board. _____

10.3 (Fill in the blank) The total number of distinct rectangles (including squares) of any size that can be formed exclusively using the internal and external grid lines is _____.

10.4 (True/False) If two diagonally opposite corner squares of the grid are removed, the remaining 62 squares can be perfectly covered without overlap by 31 standard 1×2 dominoes. _____

End of Part A

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Mock Test - Part B

Instructions: Attempt all questions. Take your time to clearly state your assumptions and justify your steps. All the best!

Question 1

15 points

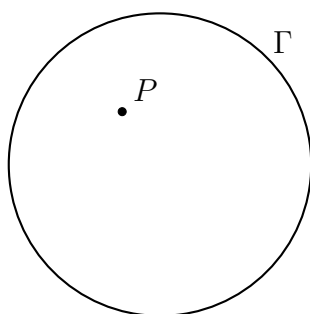
Consider a $2 \times n$ board that you want to tile using standard 1×2 dominoes, 2×1 dominoes, and 2×2 square tiles. We have one special rule: no two 2×2 squares are allowed to share an edge. Let C_n denote the number of valid tilings for a $2 \times n$ board.

- (a) Start small: Calculate C_1, C_2, C_3 , and C_4 . Using these, derive a recurrence relation for C_n . (*Hint: Try splitting into exhaustive, mutually exclusive cases based on what covers the leftmost column.*)
- (b) Using your recurrence, find a closed-form expression for C_n .
- (c) Now, let's change the rule. Let D_n be the number of tilings of the same board using the same tiles, but this time, the total number of 2×2 squares used must be an even number. Derive a recurrence relation or generating function for D_n , simplifying your final answer as much as possible.

Question 2

15 points

You are given a circle Γ and a fixed point P located strictly inside it.



- (a) Using only a straightedge, compass, and pencil, provide a clear, step-by-step construction of a chord AB of Γ such that P is exactly the midpoint of AB . Be sure to prove why your construction mathematically guarantees P is the midpoint.
- (b) Is this chord unique? Determine exactly how many such chords exist through the given point P , and explain your reasoning geometrically.

Question 3

15 points

Consider the family of continuously differentiable functions:

$$\mathcal{F} = \left\{ f \in C^1[0, 1] : f(0) = 0, \int_0^1 (f'(x))^2 dx \leq 1 \right\}$$

We are interested in maximizing the integral $M = \max_{f \in \mathcal{F}} \int_0^1 f(x) \sin x \, dx$.

- (a) First, prove that this maximum M is finite. Then, show that for every function f in our family $\mathcal{F}$, the following identity holds:

$$\int_0^1 f(x) \sin x \, dx = \int_0^1 (\cos x - \cos 1) f'(x) dx$$

- (b) Using the identity from part (a), deduce the exact value of M . What is the unique function $f \in \mathcal{F}$ that achieves this maximum?
- (c) Now, introduce a real parameter λ . Determine the maximum value of the expression $\int_0^1 f(x) dx - \lambda f(1)$ over all functions $f \in \mathcal{F}$, and express your answer explicitly as a function of λ .

Question 4

15 points

Let A be a real $n \times n$ matrix satisfying $A^3 = A$. To avoid trivial cases, assume that the matrices I , A , and A^2 are linearly independent (meaning the equation $xI + yA + zA^2 = \mathbf{0}$ holds if and only if $x = y = z = 0$). For any real numbers a, b, c , we define the matrix polynomial $T(a, b, c) = aI + bA + cA^2$.

- (a) Prove that every polynomial in A can be uniquely expressed in the form $T(a, b, c)$. Then, find all specific triples (a, b, c) that make $T(a, b, c)$ idempotent (meaning $(T(a, b, c))^2 = T(a, b, c)$) or involutory (meaning $(T(a, b, c))^2 = I$).
- (b) Find all triples (a, b, c) such that $T(a, b, c)$ is nilpotent. Also, prove that if $T(a, b, c)$ is invertible, its inverse can be written as $T(a', b', c')$, and find the explicit formulas for a' , b' , and c' in terms of a , b , and c .
- (c) Show that there is a real polynomial $q(x)$ such that $q(A) = A^2$ and $q(q(A)) = q(A)$. Can you find all such polynomials of degree 2 or less?

Question 5

15 points

Let $Q(a, b) = a^2 - ab + b^2$.

- (a) Prove that if a and b are coprime (i.e., $\gcd(a, b) = 1$), then the greatest common divisor $\gcd(a + b, Q(a, b))$ must be either 1 or 3. Use this to find all positive integer solutions to the equations $Q(a, b) = a + b$ and $Q(a, b) = 3(a + b)$.

- (b) Prove that if $\gcd(a, b) = 1$ and the square $Q(a, b)^2$ evenly divides $a^3 + b^3$, then $Q(a, b)$ can only be 1 or 3. From this, deduce all positive integer solutions where $Q(a, b)^2 | (a^3 + b^3)$.
- (c) Finally, show that for any fixed positive integer n , the equation $Q(a, b) = n$ has a strictly finite number of positive integer solutions. You can do this by proving that any such solution must satisfy $a \leq \sqrt{\frac{4n}{3}}$ and $b \leq \sqrt{\frac{4n}{3}}$.

Question 6**15 points**

Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous on the closed interval $[0, 1]$ and twice differentiable on the open interval $(0, 1)$, with the boundary values $f(0) = 0$ and $f(1) = 1$.

- (a) Show that there exists at least one point $c \in (0, 1)$ such that $f'(c) = 1$.
- (b) Now, assume the additional condition that the derivatives at the boundaries vanish: $f'(0) = f'(1) = 0$ (you may assume right and left limits respectively). Find, with rigorous proof, the strictly *largest* possible real number K such that we can guarantee there exists some $c \in (0, 1)$ satisfying $|f''(c)| \geq K$.
- (c) Prove that your bound K from part (b) is "sharp." Demonstrate this by showing that for any arbitrarily small $\epsilon > 0$, there exists a valid function $g(x)$ satisfying all the given conditions where $|g''(x)| < K + \epsilon$ for all $x \in (0, 1)$.

End of Part B